

Matrices, Vectors & Complex Numbers



Matrix arithmetic

- 1 Matrix A has rows $(2, 3)$ and $(-1, 4)$; matrix B has rows $(5, -2)$ and $(1, 3)$. Find the rows of $A + B$.
 - 2 Find $3A$ for matrix A with rows $(1, -2)$ and $(4, 0)$.
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Matrix multiplication

- 3 Matrix A has rows $(1, 2)$ and $(3, 4)$; matrix B has rows $(5, 6)$ and $(7, 8)$. Find AB .
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Determinants and inverses of 2×2 matrices

- 4 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 5)$.
 - 5 Find the inverse of the matrix M with rows $(2, 1)$ and $(5, 3)$.
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Vectors: magnitude and direction

- 6 Find the magnitude of the vector $\mathbf{v} = (3, 4)$.
 - 7 Given $\mathbf{a} = (2, -5)$ and $\mathbf{b} = (-3, 7)$, find $\mathbf{a} + \mathbf{b}$ and $2\mathbf{a} - \mathbf{b}$.
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The dot product

- 8 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 2)$ and $\mathbf{b} = (-1, 4)$, and determine whether the vectors are perpendicular.
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Complex number arithmetic

- 9 Simplify $(4 + 3i) + (2 - 5i)$.
 - 10 Simplify $(3 + 2i)(1 - 4i)$.
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Complex conjugates and division

- 11 Find the complex conjugate of $z = 6 - 5i$, and compute $z\bar{z}$.
- 12 Simplify $\frac{5+i}{2-i}$ by multiplying by the conjugate of the denominator.
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Solving quadratics with complex solutions

- 13 Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.
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Combining matrices and systems of equations

- 14 This question has two parts. The system $3x + 2y = 16$ and $x + 4y = 18$ can be written as a matrix equation $M\mathbf{v} = \mathbf{c}$, where M has rows $(3, 2)$ and $(1, 4)$. (a) Find $\det(M)$ and M^{-1} . (b) Use M^{-1} to solve the system for x and y .
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Powers of i and simplifying complex expressions

- 15 Simplify i^{17} , using the fact that powers of i cycle with period 4.
- 16 Simplify $(2i)^3$.
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Vector direction and unit vectors

- 17 Find a unit vector in the same direction as $\mathbf{v} = (6, -8)$.
- 18 Find the angle a vector $\mathbf{v} = (3, 3)$ makes with the positive x -axis.
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Identity matrix and matrix equations

- 19 Explain why $MI = M$ for any matrix M and identity matrix I of matching size, and verify this for M with rows $(2, 3)$ and $(1, 4)$ multiplied by the 2×2 identity matrix (rows $(1, 0)$ and $(0, 1)$).
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A multi-part complex number problem

- 20 This question has two parts. Given $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. (a) Find $|z_1|$ (the modulus of z_1). (b) Find $z_1 \cdot z_2$ and its modulus, then verify that $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$.

Scalar multiplication of matrices and vectors combined

21 Given vectors $\mathbf{u} = (4, -3)$ and $\mathbf{v} = (-2, 5)$, find $3\mathbf{u} - 2\mathbf{v}$.